

Oppgave 1

$$AC(Q) = \frac{TC(Q)}{Q} = \frac{3000}{Q} + 1,2Q^{0,1}$$

$$\frac{dAC}{dQ}(Q) = -\frac{3000}{Q^2} + 0,12Q^{-0,9} = 0 \Rightarrow Q^{1,1} = \frac{3000}{0,12} \Rightarrow$$

$$Q = \left(\frac{3000}{0,12}\right)^{1/1,1} = \mathbf{9956,972 = 9957}$$

Oppgave 2

For monopol er $Q_P = Q_D = Q$

$$TR(Q) = Q \cdot P(Q) = 50Q - 1,2Q^{1,5} \Rightarrow MR(Q) = \frac{dTR}{dQ}(Q) = 50 - 1,8Q^{0,5}$$

$$TC(Q) = FC + Q^{1,5} \Rightarrow MC(Q) = \frac{dTC}{dQ}(Q) = 1,5Q^{0,5}$$

Profitten maksimeres når $MR(Q) = MC(Q) \Rightarrow$

$$50 - 1,8Q^{0,5} = 1,5Q^{0,5} \Rightarrow Q^{0,5} = \frac{50}{3,3} \Rightarrow Q^* = \left(\frac{50}{3,3}\right)^2 = 229,5684$$

Profitten er $\Pi(Q) = TR(Q) - TC(Q) \Rightarrow \Pi(Q^*) = 1500 \Rightarrow$

$$7304,449 - (FC + 3478,309) = 1500 \Rightarrow FC = \mathbf{2326,14 \Rightarrow 2326}$$

Oppgave 3

Et perfekt konkurranseutsatt innebærer at $P(Q) = MC(Q)$ dvs. vi har den opprinnelige løsningen

$$P(Q) = 200 - 3Q = 110 \Rightarrow Q = 30$$

$$KO_{org} = \int_0^{30} (200 - 3q - 110) dq = \left[90q - \frac{3q^2}{2} \right]_0^{30} = 90 \cdot 30 - \frac{3 \cdot 30^2}{2} = 1350$$

og den nye løsningen

$$P(Q) = 200 - 3Q = 90 \Rightarrow Q = 36,667$$

$$KO_{ny} = \int_0^{36,667} (200 - 3q - 90) dq = \left[110q - \frac{3q^2}{2} \right]_0^{36,667} = 2016,6667$$

Dvs. en økning av konsumentoverskuddet med

$$\Delta_{KO} = KO_{ny} - KO_{org} = 2016,6667 - 1350 = \mathbf{666,6667 \Rightarrow 667}$$

Oppgave 4

$$Kortsiktig gjeld = 150 \cdot (1 - 2/3) = 50$$

$$Omløpsmidler = 150$$

$$Mest LOM = 150 \cdot (1 - 2/5) = 90$$

$$Ubenyttet kassakreditt = 1,01$$

$$\text{Likviditetsgrad 1} = \frac{150 - x + 1,01}{50} > 2 \Rightarrow x < 150 + 1,01 - 2 \cdot 50 = 51,01$$

$$\text{Likviditetsgrad 2} = \frac{90 - x + 1,01}{50} > 1 \Rightarrow x < 90 + 1,01 - 50 = 41,01$$

Maks **41,01** $\Rightarrow$ **41** kan betales i utbytte til eierne for både likviditetsgrad 1 og 2 skal oppfylle tommelfingerreglene for god likviditet.

Oppgave 5

$$Gjeldsgrad = \frac{D}{E} = 1,5 \Rightarrow D = 1,5E$$

$$Gjeldsandel = \frac{D}{D+E} = \frac{1,5E}{1,5E+E} = \frac{1,5}{2,5} = \mathbf{0,60}$$

Oppgave 6

$$SM_{org} = 1500 - \frac{FC}{p_{org}-VC_{st}} = 1500 - \frac{10\,000}{30-22} = 250$$

$$SM_{ny} = Q - \frac{FC}{p_{ny}-VC_{st}} = Q - \frac{10\,000}{30-0,9-22} = Q - 2000 = 250 \Rightarrow$$

$$Q = 2250 = 1500 + \mathbf{750}$$

Oppgave 7

$$R_{tot,f} = \frac{Driftsresultat+finansinntekter}{Tot.kapital}$$

$$0,1200 = \frac{10\,000+2000}{Tot.kapital} \Rightarrow Tot.kapital = 100\,000$$

$$\frac{10\,000+780 \cdot (80-0,02)+2000}{100\,000} = 0,13248 = 0,1200 + \mathbf{0,01248} \Rightarrow \mathbf{0,0125}$$

Oppgave 8

$$\frac{200}{(1+r)^2} = 180 \Rightarrow r = \left(\frac{200}{180}\right)^{1/2} - 1 = \mathbf{0,054093} \Rightarrow \mathbf{0,0541}$$

Oppgave 9

$$-1800 + \frac{400}{1,0789^1} + \frac{400}{1,0789^2} + \frac{400}{1,0789^3} + \frac{400}{1,0789^4} + \frac{400}{1,0789^5} + \frac{200}{1,0789^6} + \frac{200+100}{1,0789^7} = 104,83$$

$$-1800 + \frac{400}{1,0969^1} + \frac{400}{1,0969^2} + \frac{400}{1,0969^3} + \frac{400}{1,0969^4} + \frac{400}{1,0969^5} + \frac{200}{1,0969^6} + \frac{200+100}{1,0969^7} = 0,24$$
$$\Rightarrow \mathbf{9,69\%}$$

Oppgave 10

$$\frac{NNV_A}{G_A} = \left(-1000 + 250 \frac{1-1,06^{-5}}{0,06} + \frac{150}{1,06^5}\right)/1000 = 0,1651797$$

$$\frac{NNV_B}{G_B} = \left(-800 + x \frac{1-1,06^{-5}}{0,06} + \frac{200}{1,06^5}\right)/800 = 0,1651797 \Rightarrow$$

$$x = \left(800 \cdot 0,1651797 + 800 - \frac{200}{1,06^5}\right)/\frac{1-1,06^{-5}}{0,06} = \mathbf{185,8083} \Rightarrow \mathbf{186}$$

Oppgave 11

Vekting massebasert allokering:

$$m_{\text{total}} = m_A + m_B + m_C$$

$$m_{\text{total}} = 300\text{kg} + 600\text{kg} = 900\text{kg},$$

$$y_A = \frac{m_A}{m_{\text{total}}}$$

$$y_A = \frac{300\text{kg}}{900\text{kg}} \approx \underline{0,3333}.$$

Allokerte utslipp produkt A:

$$h_A = y_A \cdot h$$

$$h_A \approx 0,3333 \cdot 60\text{kg CO}_2 = \underline{\underline{20\text{kg CO}_2}}.$$

Oppgave 12

Energiinnhold:

$$E_{\text{total}} = E_A + E_B + E_C$$

$$E_A = m_A \cdot e_A = 30\text{kg} \cdot 8\text{MJ/kg} = \underline{240\text{MJ}},$$

$$E_B = m_B \cdot e_B = 90\text{kg} \cdot 13\text{MJ/kg} = \underline{1170\text{MJ}},$$

$$E_C = m_C \cdot e_C = 10\text{kg} \cdot 6\text{MJ/kg} = \underline{60\text{MJ}},$$

$$E_{\text{total}} = 240\text{MJ} + 1170\text{MJ} + 60\text{MJ} = \underline{1470\text{MJ}}.$$

Vekting økonomisk allokering:

$$y_A = \frac{E_A}{E_{\text{total}}}$$

$$y_A = \frac{240\text{MJ}}{1170\text{MJ}} \approx \underline{0,1633}.$$

Allokerte utslipp produkt A:

$$h_A = y_A \cdot h$$

$$h_A \approx 0,1633 \cdot 86\text{kg CO}_2\text{eq} \approx 14,04\text{kg CO}_2\text{eq} \approx \underline{\underline{14\text{kg CO}_2\text{eq}}}.$$

Oppgave 13

Totalt klimagassutslipp:

$$h_i = \sum_j \phi_{ij} \cdot r_j$$

$$h_{\text{GWP}} = 1\text{kgCO}_2\text{eq/kg} \cdot 40\text{kg} + 30\text{kgCO}_2\text{eq/kg} \cdot 9,0\text{kg}$$

$$+ 1300\text{kgCO}_2\text{eq/kg} \cdot 0,50\text{kg} + 23500\text{kgCO}_2\text{eq/kg} \cdot 0,040\text{kg}$$

$$h_{\text{GWP}} = \underline{\underline{1900\text{kgCO}_2\text{eq}}}.$$

Oppgave 14

Teknologimatrise:

	A	B	C
Elektrisitet (kWh)	10	-20	-8
Plast (kg)	0	1	-2,4
Flasker (stk)	0	0	100

Manglende verdier:

10	-20 A	-8 B
0 C	1 D	-2,4
0 E	0	100

Oppgave 15

Utslipp SO₂ prosess A: 200 g

Produksjon drivstoff prosess A: 160 L

SO₂-utslipp per L drivstoff levert fra prosess A: $200 \text{ g}/160 \text{ L} = \underline{1,25 \text{ g/L}}$.

Oppgave 16

Globalt oppvarmingspotensial N₂O:

$$h_i = \sum_j \phi_{ij} \cdot r_j$$

$$h_{\text{GWP}, \text{N}_2\text{O}} = 273,0 \text{ kgCO}_2 \text{ eq/kg} \cdot 5,00 \text{ kg}$$

$$h_{\text{GWP}, \text{N}_2\text{O}} = \underline{1365 \text{ kgCO}_2 \text{ eq.}}$$

Globalt oppvarmingspotensial totalt:

$$h_i = \sum_j \phi_{ij} \cdot r_j$$

$$h_{\text{GWP}} = 1,0 \text{ kgCO}_2 \text{ eq/kg} \cdot 25,00 \text{ kg} + 30,0 \text{ kgCO}_2 \text{ eq/kg} \cdot 10,00 \text{ kg} \\ + 273,0 \text{ kgCO}_2 \text{ eq/kg} \cdot 5,00 \text{ kg} + 23500,0 \text{ kgCO}_2 \text{ eq/kg} \cdot 0,20 \text{ kg}$$

$$h_{\text{GWP}} = \underline{6390 \text{ kgCO}_2 \text{ eq.}}$$

Andel N₂O:

$$y_{\text{N}_2\text{O}} = \frac{h_{\text{GWP}, \text{N}_2\text{O}}}{h_{\text{GWP}}}$$

$$y_{\text{N}_2\text{O}} = \frac{1365 \text{ kgCO}_2 \text{ eq}}{6390 \text{ kgCO}_2 \text{ eq}}$$

$$y_{\text{N}_2\text{O}} \approx 0,2136 \approx 0,21 = \underline{\underline{21\%}}$$

Oppgave 17

Materialbalanse:

$$X_{\text{ut}} = X_{\text{inn}} - X_{\text{akkumulert}}$$

$$X_4 + X_5 = X_1 + X_2 + X_3 - X_{\text{akkumulert}}$$

$$X_5 = X_1 + X_2 + X_3 - X_{\text{akkumulert}} - X_4$$

$$X_5 = 1000 \text{ kg} + 2000 \text{ kg} + 3000 \text{ kg} - 1000 \text{ kg} - 2000 \text{ kg}$$

$$X_5 = \underline{\underline{3000 \text{ kg.}}}$$

Oppgave 18

Materialbalanse (stasjonær tilstand):

- Totalt system:

$$x_{01} = x_{10} + x_{30}$$

$$x_{01} = 25 \text{ kg} + 70 \text{ kg}$$

$$x_{01} = \underline{\underline{95 \text{ kg.}}}$$

Kan også fastsettes fra materialbalanse for alle prosessene.

Oppgave 19

Allokeringskoeffisient for stoffstrøm X1:

$$k_i = \frac{X_i}{\sum_j X_j}$$

$$k_{x1} = \frac{X1}{X1 + X2} = \frac{X1}{X3 + X4 + X5}$$

$$k_{x1} = \frac{4000\text{kg}}{4000\text{kg} + 6000\text{kg}} = \underline{\underline{0,40.}}$$

Oppgave 20

Koeffisientmatrise og vektor konstanter:

$$A = \begin{bmatrix} 1 & -1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -0,10 & 0 & 1 \\ -0,05 & 1 & 0 & 0 & -0,05 \end{bmatrix} \quad x = \begin{bmatrix} x_{01} \\ x_{10} \\ x_{12} \\ x_{20} \\ x_{21} \end{bmatrix} \quad y = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 0 \\ 0 \end{bmatrix}$$

Feil i oppsatt koeffisientmatrise og vektor konstanter:

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & -1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & -0,10 & 0 & 1 \\ -0,05 & 1 & 0 & 0 & 0 \end{bmatrix} \quad x = \begin{bmatrix} x_{01} \\ x_{10} \\ x_{12} \\ x_{20} \\ x_{21} \end{bmatrix} \quad y = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 0 \\ 0 \end{bmatrix}$$